UK Data Dashboard Project

UK Local Authority Greenhouse Gas Emissions - Interactive Excel Dashboard

SkillVersed Data Analytics | Excel Level 2

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Non-AI domain project using a UK public data topic | 7,000-record cleaned dataset

Project summary. This report explains the design and development of an interactive Excel dashboard built from a 7,000-record working dataset about UK local authority greenhouse gas emissions. The dashboard is intended for non-technical users and is supported by cleaned data, PivotTables, slicers, KPI cards, charts, a source sheet, and a data dictionary.

Submission Deliverables Included

Deliverable	Description
Excel dashboard workbook	ShraddhaYadav_UKDashboard_7000_REAL_Pivots_Slicers.xlsx - cleaned dataset, active PivotTables, slicers, KPIs, charts, source and data dictionary sheets.
Project report	This Word report and exported PDF version covering dataset selection, evaluation, audience, analytical perspective, dashboard design and testing.
ZIP folder	A single submission folder containing the dashboard workbook and report files.

1. Dataset Selection

For this project I selected a UK public-interest dataset based on local authority and regional greenhouse gas emissions. The topic is suitable for the UK Data Dashboard Project because it is not about one company or brand; it represents a wider public and environmental issue that affects councils, regions, policy teams and residents across the UK.

The dataset used in the dashboard is a cleaned 7,000-record working extract. The working file keeps the important structure needed for analysis: geography, year, emissions sector, sub-sector, greenhouse gas type and numeric measures such as territorial emissions, scope-of-influence emissions, population, area, emissions per capita and emissions density.

I chose this dataset because greenhouse gas emissions and net-zero planning are important public policy topics. The data can be analysed by place, by time period, by sector and by gas, which makes it very suitable for an interactive Excel dashboard with PivotTables and slicers.

Source note: The original public data source is the UK government greenhouse gas emissions statistics published by the Department for Energy Security and Net Zero (DESNZ) on GOV.UK. The workbook Source sheet records the publisher, licence, period and official source link for traceability.

2. Dataset Evaluation and Documentation

Before designing the dashboard, I reviewed the dataset fields and prepared a clean structure suitable for PivotTables. The data was checked so the dashboard could calculate totals, averages, counts and ratios correctly.

Group	Columns / Examples	Purpose in the dashboard
Geography	Country, Region, Local Authority, ONS codes	Allows comparison between UK nations, regions and local authority areas.
Time	Calendar Year	Supports trend analysis and year filtering.
Category	LA GHG Sector, Sub-sector, Greenhouse gas	Shows where emissions come from and lets the user filter by activity type and gas.
Measures	Territorial emissions, scope emissions, population, area, per-capita and density values	Provides numeric values for KPIs, charts and comparisons.
Banding	Emissions band	Groups records by size for distribution analysis.

The cleaned working dataset contains 7,000 records and 19 columns. Each record represents one combination of area, year, sector, sub-sector and greenhouse gas. This level of detail is useful because it allows the same dashboard to answer several questions without changing the underlying workbook.

- No blank rows were kept in the working dataset.
- Key fields such as country, region, year, sector and greenhouse gas were populated.
- Numeric columns were prepared as numbers so PivotTables and KPI formulas can aggregate them.
- The workbook includes a Source sheet and a Data Dictionary sheet so a user can understand where the data came from and what each column means.

3. End-Consumer Identification

The dashboard was designed for a non-technical audience. The main users are people who need quick insight from UK emissions data without building formulas or PivotTables themselves.

- Council officers and sustainability teams who need to understand emissions by area and sector.
- Local councillors and policy decision-makers who want headline figures and clear comparisons.
- Students, researchers and members of the public who want to explore UK emissions in a simple visual way.

The primary target audience is council and sustainability decision-makers. They are the most relevant users because they need clear evidence for planning, reporting and communicating emissions priorities. For that reason, the dashboard uses plain labels, KPI cards, slicers and simple chart titles rather than technical modelling language.

4. Perspective and Analysis Definition

The analytical perspective for the dashboard is: Where do the UK local authority greenhouse gas emissions come from, and how do they change across geography, sector, gas and year?

This perspective was chosen because it gives the user a practical way to compare emissions by location, sector and gas type. It also supports trend analysis over time, which helps users see whether emissions are rising or falling in the selected area or category.

KPI	What it shows
Total emissions	Total territorial emissions for the current filter selection.
Records in view	Number of records included after slicer filtering.
Scope emissions	Total scope-of-influence emissions for the selected data.
CO2 emissions	Total emissions for carbon dioxide within the current selection.
Average emissions per record	Average territorial emissions value per row.
CO2 share of total	Percentage of total emissions made up by CO2.

The main dashboard filters are Country, Region, Calendar Year, Sector and Greenhouse gas. These dimensions were selected because they let the user ask common questions such as which region has the highest emissions, which sector contributes most, or how a specific gas changes over time.

5. Dashboard Design and Development

5.1 Layout and structure

The dashboard is arranged from top to bottom. The top section contains the title and KPI cards, the middle section contains slicers, and the lower section contains charts. This layout makes the page easy to read because users first see the headline numbers and then explore the detail through charts and filters.

5.2 PivotTables, slicers and KPI formulas

The workbook is built using active PivotTables as the analysis engine. The PivotTables summarise the cleaned 7,000-record data table, and the dashboard charts are based on those pivot summaries. The slicers are connected to the PivotTables so that selecting a filter changes the PivotTables, charts and KPI cards together.

The KPI cards use PivotTable-driven formulas such as GETPIVOTDATA or linked pivot totals. This makes the KPI values active rather than manually typed. When the slicers change the PivotTable results, the KPI values update as well.

5.3 Power Query position

Power Query is not required as the main control for this dashboard because the working data has already been cleaned before the PivotTables were created. The correct dashboard flow is cleaned data table to PivotTables, PivotTables to slicers, and PivotTables to dashboard charts and KPIs. Power Query can be added later only if the trainer specifically wants to see the cleaning workflow inside Excel.

5.4 Chart choices

- Territorial emissions by region - bar chart for regional comparison.
- Emissions by sector - column chart to compare sector totals.
- Emissions trend by year - line chart to show change over time.
- Emissions by greenhouse gas - doughnut chart to show proportional contribution.
- Emissions by country - bar chart to compare UK nations.
- Records by emissions band - column chart showing distribution by emissions size.
- Scope-of-influence emissions by region - bar chart for comparison with territorial emissions.
- Average emissions per capita by country - bar chart for fairer population-based comparison.

5.5 User experience and visual design

The design uses clear headings, readable number formatting, consistent chart titles and simple slicers. This is important for a non-technical audience because users should be able to explore the dashboard by clicking filters rather than editing formulas or looking at raw data.

6. Testing and Refresh Process

The dashboard was tested by refreshing the workbook and checking that slicers changed the PivotTables, charts and KPI values. The main refresh process in Excel is Data > Refresh All. If the dataset is updated later, the user should refresh the workbook and then check the slicer report connections for the PivotTables.

- The dataset table covers 7,000 records.
- PivotTables are used as the active analysis layer.
- Slicers are connected to PivotTables rather than directly to Power Query.
- KPI values are linked to pivot results so they respond to filter selections.
- The workbook includes source and data dictionary documentation.

7. Conclusion

The finished workbook meets the project goal by using a UK public-domain topic, a cleaned 7,000-record dataset, active PivotTables, slicers, KPI cards and charts to create an interactive dashboard for a non-technical audience. The design allows users to explore UK greenhouse gas emissions by country, region, year, sector and gas type without needing to work directly with the raw data.